

Computer Vision 2: Interview-Focused Guide

This guide covers advanced computer vision topics for ML placements, AI engineer roles, research internships, and projects. Each topic moves from basics to interview depth with formulas, code, evaluation advice, mistakes, limitations, and project ideas.

Batch Normalization

1. Overview

Batch normalization normalizes hidden activations inside a neural network using mini-batch statistics, then learns a scale and shift. It is used to stabilize deep CNN/MLP training, improve gradient flow, allow larger learning rates, and reduce sensitivity to initialization.

2. Intuition

A layer trains better when the distribution of its inputs is not wildly changing. BatchNorm keeps activations in a predictable range, like giving every layer a cleaner signal before it learns its own transformation.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Mini-batch statistics

- What it means: Mean and variance are computed from the current mini-batch.
- Why it matters: It controls whether Batch Normalization works correctly in training, evaluation, or deployment.
- Simple example: A channel in a CNN is normalized across batch and spatial positions.
- Common interview angle: Why does BatchNorm behave poorly with very small batches?

Affine parameters

- What it means: Learnable gamma and beta restore representational freedom.
- Why it matters: It controls whether Batch Normalization works correctly in training, evaluation, or deployment.
- Simple example: If unit variance is not ideal, gamma learns the useful scale.
- Common interview angle: Why does normalization not reduce model capacity?

Running statistics

- What it means: Moving averages are saved for inference.
- Why it matters: It controls whether Batch Normalization works correctly in training, evaluation, or deployment.
- Simple example: One test image should not define its own batch statistics.
- Common interview angle: Why must `model.eval()` be used?

Placement

- What it means: Often Conv -> BatchNorm -> ReLU.
- Why it matters: It controls whether Batch Normalization works correctly in training, evaluation, or deployment.
- Simple example: ResNet blocks commonly normalize convolution output before activation.
- Common interview angle: Where do you place BatchNorm and why?

5. Algorithm / Working Process

- Input: Layer activations, commonly shaped $N \times C \times H \times W$ in CNNs.
- Processing steps:
 1. Receive activations from previous layer.
 2. Compute batch mean and variance.
 3. Normalize activations.
 4. Apply gamma and beta.
 5. Update running statistics during training.
 6. Use running statistics during inference.
- Output: Normalized and re-scaled activations with the same shape.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $\mu_B = (1/m) \sum_i x_i$
- $\sigma_B^2 = (1/m) \sum_i (x_i - \mu_B)^2$
- $x_{\text{hat}_i} = (x_i - \mu_B) / \sqrt{\sigma_B^2 + \epsilon}$
- $y_i = \gamma x_{\text{hat}_i} + \beta$

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
import torch
from torch import nn

model = nn.Sequential(
    nn.Conv2d(3, 32, 3, padding=1, bias=False),
    nn.BatchNorm2d(32),
```

```
nn.ReLU(inplace=True),
nn.AdaptiveAvgPool2d(1),
nn.Flatten(),
nn.Linear(32, 10),
)

x = torch.randn(8, 3, 64, 64)
model.train()
loss = nn.CrossEntropyLoss()(model(x), torch.randint(0, 10, (8,)))
loss.backward()

model.eval()
with torch.no_grad():
    pred = model(x).argmax(dim=1)
```

8. Code Explanation

- The imports use the smallest common library for practice.
- The model block shows either a pretrained model or a tiny runnable version.
- The input block demonstrates the expected shape or file input.
- The output block prints the object you must understand in interviews: logits, boxes, masks, embeddings, generated image, or coordinates.
- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: validation accuracy, training stability, loss smoothness, generalization gap.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- CNN image classifiers
- detection backbones
- segmentation encoders

- transfer learning
- deep residual networks

12. Common Mistakes

- forgetting `model.eval()`
- using tiny batches without GroupNorm/LayerNorm
- freezing weights but mishandling BN stats
- thinking BN is the same as input scaling
- expecting BN to replace regularization

13. Edge Cases / Limitations

- unstable with tiny batches
- train/inference behavior differs
- can leak batch information
- less common in modern Transformers

14. Variations

Item	Details
LayerNorm	per-sample feature normalization, key for Transformers
GroupNorm	batch-size independent, useful for detection and segmentation
InstanceNorm	common in style transfer
SyncBatchNorm	synchronizes stats across GPUs

15. Related Topics

- BatchNorm vs LayerNorm: batch stats vs per-sample stats
- BatchNorm vs Dropout: stabilization vs random regularization
- BatchNorm and ResNet: helps train very deep CNNs

16. Interview Questions

1. **What problem does it solve?** Answer: It solves the core task described in the overview and makes that task trainable, scalable, or deployable in real computer-vision systems.
2. **What is the input and output?** Answer: Input and output are listed in the working process; always answer with tensor/object shapes when possible.
3. **What is the main training signal?** Answer: The training signal is the relevant loss: cross-entropy, box loss, mask loss, contrastive loss, denoising loss, reconstruction loss, or language-modeling loss.
4. **Which metric should be used?** Answer: Choose the metric that matches the output, not generic accuracy.
5. **How do you debug bad results?** Answer: Visualize predictions, inspect labels, overfit a tiny batch, verify preprocessing, and check train/validation curves.
6. **What causes overfitting?** Answer: Small data, noisy labels, weak augmentation, too-large models, and validation leakage.

7. **What hyperparameters matter most?** Answer: Learning rate, batch size, resolution, model size, thresholds, augmentation, scheduler, and loss weights.
8. **What is the most common interview trap?** Answer: Only describing the architecture and forgetting loss, metrics, inference post-processing, and failure modes.
9. **How do you improve performance?** Answer: Clean data first, use pretrained weights, tune preprocessing and thresholds, then scale or modify architecture.
10. **How do you deploy it?** Answer: Export the model, reproduce preprocessing, benchmark latency/memory, monitor drift, and test on real edge cases.

17. Practice Tasks

- Small coding task: run the provided snippet and explain every tensor/output shape.
- Dataset-based project: train or evaluate on a tiny public/custom dataset.
- Experiment idea: compare pretrained vs randomly initialized behavior.
- Debugging task: break preprocessing or labels and identify the metric/visual symptom.
- Extension idea: add a visualization script that saves predictions for review.

18. Project Ideas

Batch Normalization Mini Lab

- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
- Dataset suggestion: COCO subset, Oxford Pets, Cityscapes subset, medical masks, satellite images, or custom data
- Resume value: Shows implementation and evaluation ability.

Batch Normalization Error Analysis Dashboard

- What it does: Save predictions, metrics, and failure examples.
- Tech stack: Python, Streamlit, pandas, OpenCV
- Dataset suggestion: A validation set with diverse hard examples
- Resume value: Strong resume signal because it shows engineering maturity.

Batch Normalization Domain Adaptation Demo

- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
- Resume value: Shows transfer learning and practical ML judgment.

19. Quick Revision

- Key idea: Batch normalization normalizes hidden activations inside a neural network using mini-batch statistics, then learns a scale and shift.
- Main formula: $\mu_B = (1/m) \sum_i x_i$.
- When to use: CNN image classifiers.
- Important metrics: validation accuracy, training stability, loss smoothness.
- Common traps: forgetting `model.eval()`; using tiny batches without GroupNorm/LayerNorm.

- Interview one-liner: Batch Normalization is a normalization layer used for CNN image classifiers and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	Batch Normalization is a normalization layer.
Input/output	Layer activations, commonly shaped $N \times C \times H \times W$ in CNNs. -> Normalized and re-scaled activations with the same shape.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	validation accuracy, training stability, loss smoothness, generalization gap
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	CNN image classifiers, detection backbones, segmentation encoders

YOLO

1. Overview

YOLO, You Only Look Once, predicts bounding boxes and class probabilities in a single forward pass. It is used when real-time detection matters: traffic monitoring, drones, robotics, retail analytics, sports tracking, and edge AI.

2. Intuition

YOLO looks at the whole image once and directly says what objects exist and where they are. It avoids a separate proposal stage, so it is fast.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Dense prediction

- What it means: The detector predicts boxes across feature-map locations.
- Why it matters: It controls whether YOLO works correctly in training, evaluation, or deployment.
- Simple example: Every location can predict nearby objects.
- Common interview angle: Why is YOLO faster than two-stage detectors?

Objectness

- What it means: A score measuring whether a predicted box contains an object.
- Why it matters: It controls whether YOLO works correctly in training, evaluation, or deployment.
- Simple example: Low-objectness boxes are removed before NMS.
- Common interview angle: What is objectness?

Box regression

- What it means: The model predicts center, width, and height or equivalent offsets.
- Why it matters: It controls whether YOLO works correctly in training, evaluation, or deployment.
- Simple example: A car is represented by x_1, y_1, x_2, y_2 after decoding.
- Common interview angle: How is localization trained?

NMS

- What it means: Non-maximum suppression removes duplicate boxes.
- Why it matters: It controls whether YOLO works correctly in training, evaluation, or deployment.
- Simple example: Ten person boxes become one final person detection.
- Common interview angle: When can NMS fail?

5. Algorithm / Working Process

- Input: An image resized to a fixed resolution.
- Processing steps:
 1. Resize and normalize image.
 2. Backbone extracts features.
 3. Neck fuses multi-scale features.
 4. Head predicts boxes, objectness, and classes.
 5. Decode boxes to image coordinates.
 6. Filter confidence and apply NMS.
- Output: Bounding boxes, confidence scores, and class labels.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $\text{IoU} = \text{area}(\text{intersection}) / \text{area}(\text{union})$
- $\text{score} = \text{objectness} \times \text{class_probability}$
- $\text{loss} = \text{box_loss} + \text{objectness_loss} + \text{class_loss}$
- mAP averages precision over recall and classes

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
# pip install ultralytics
from ultralytics import YOLO

model = YOLO("yolov8n.pt")
model.train(data="coco8.yaml", epochs=5, imgsz=640, batch=8)

results = model("image.jpg", conf=0.25, iou=0.45)
for r in results:
    print(r.boxes.xyxy, r.boxes.conf, r.boxes.cls)
```

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- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: mAP@0.5, mAP@0.5:0.95, precision, recall, FPS.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- vehicle detection

- people counting
- defect detection
- drone detection
- sports analytics

12. Common Mistakes

- wrong box format
- bad confidence threshold
- using accuracy instead of mAP
- not visualizing labels
- ignoring small-object performance

13. Edge Cases / Limitations

- crowded scenes challenge NMS
- small objects can be missed
- domain shift hurts
- boxes are not masks
- fast variants may lose accuracy

14. Variations

Item	Details
YOLOv3/v5	popular anchor-based baselines
YOLOv8 style	modern practical detector family
Tiny YOLO	edge deployment
YOLO-seg/pose	masks or keypoints

15. Related Topics

- YOLO vs Faster R-CNN: speed vs proposal refinement
- YOLO vs SSD: both one-stage dense detectors
- YOLO plus tracking: detections feed SORT/DeepSORT/ByteTrack

16. Interview Questions

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2. **What is the input and output?** Answer: Input and output are listed in the working process; always answer with tensor/object shapes when possible.
3. **What is the main training signal?** Answer: The training signal is the relevant loss: cross-entropy, box loss, mask loss, contrastive loss, denoising loss, reconstruction loss, or language-modeling loss.
4. **Which metric should be used?** Answer: Choose the metric that matches the output, not generic accuracy.

5. **How do you debug bad results?** Answer: Visualize predictions, inspect labels, overfit a tiny batch, verify preprocessing, and check train/validation curves.
6. **What causes overfitting?** Answer: Small data, noisy labels, weak augmentation, too-large models, and validation leakage.
7. **What hyperparameters matter most?** Answer: Learning rate, batch size, resolution, model size, thresholds, augmentation, scheduler, and loss weights.
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17. Practice Tasks

- Small coding task: run the provided snippet and explain every tensor/output shape.
- Dataset-based project: train or evaluate on a tiny public/custom dataset.
- Experiment idea: compare pretrained vs randomly initialized behavior.
- Debugging task: break preprocessing or labels and identify the metric/visual symptom.
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18. Project Ideas

YOLO Mini Lab

- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
- Dataset suggestion: COCO subset, Oxford Pets, Cityscapes subset, medical masks, satellite images, or custom data
- Resume value: Shows implementation and evaluation ability.

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- What it does: Save predictions, metrics, and failure examples.
- Tech stack: Python, Streamlit, pandas, OpenCV
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YOLO Domain Adaptation Demo

- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
- Resume value: Shows transfer learning and practical ML judgment.

19. Quick Revision

- Key idea: YOLO, You Only Look Once, predicts bounding boxes and class probabilities in a single forward pass.

- Main formula: $\text{IoU} = \text{area}(\text{intersection}) / \text{area}(\text{union})$.
- When to use: vehicle detection.
- Important metrics: mAP@0.5 , mAP@0.5:0.95 , precision.
- Common traps: wrong box format; bad confidence threshold.
- Interview one-liner: YOLO is a one-stage object detector used for vehicle detection and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	YOLO is a one-stage object detector.
Input/output	An image resized to a fixed resolution. -> Bounding boxes, confidence scores, and class labels.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	mAP@0.5 , mAP@0.5:0.95 , precision, recall
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	vehicle detection, people counting, defect detection

Faster R-CNN

1. Overview

Faster R-CNN detects objects using a Region Proposal Network followed by a second-stage classifier and box regressor. It is a classic high-accuracy detector and a core placement interview architecture.

2. Intuition

First find promising regions, then inspect each region carefully. The first stage asks where objects might be; the second asks what object it is and tightens the box.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

RPN

- What it means: Region Proposal Network predicts object-like candidate boxes.
- Why it matters: It controls whether Faster R-CNN works correctly in training, evaluation, or deployment.
- Simple example: It proposes regions before final classification.
- Common interview angle: What made Faster R-CNN faster than Fast R-CNN?

Anchors

- What it means: Predefined boxes at different scales and aspect ratios.
- Why it matters: It controls whether Faster R-CNN works correctly in training, evaluation, or deployment.
- Simple example: Tall, wide, and square anchors cover likely objects.
- Common interview angle: How are positives assigned by IoU?

ROI Align

- What it means: Converts variable proposals into fixed-size features.
- Why it matters: It controls whether Faster R-CNN works correctly in training, evaluation, or deployment.
- Simple example: A 120x80 proposal becomes a 7x7 feature grid.
- Common interview angle: Why is ROI Align better than ROI Pooling?

Two-stage refinement

- What it means: The second head classifies and refines proposals.
- Why it matters: It controls whether Faster R-CNN works correctly in training, evaluation, or deployment.
- Simple example: A rough dog proposal becomes a tight dog box.
- Common interview angle: Why is it accurate but slower?

5. Algorithm / Working Process

- Input: Image tensor plus ground-truth boxes/classes during training.
- Processing steps:
 1. Backbone extracts feature maps.
 2. RPN predicts proposal objectness and offsets.
 3. Top proposals are selected with NMS.
 4. ROI Align crops proposal features.
 5. Detection head predicts class and box refinement.
 6. Loss combines RPN and detector objectives.
- Output: Final boxes, class labels, and confidence scores.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $t_x = (x - x_a) / w_a$, $t_y = (y - y_a) / h_a$

- $t_w = \log(w/w_a)$, $t_h = \log(h/h_a)$
- $\text{loss} = L_{\text{cls}} + \lambda L_{\text{reg}}$
- SmoothL1 is common for box regression

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
import torch
from torchvision.models.detection import fasterrcnn_resnet50_fpn

model = fasterrcnn_resnet50_fpn(weights="DEFAULT").eval()
image = torch.rand(3, 480, 640)

with torch.no_grad():
    out = model([image])[0]

keep = out["scores"] > 0.7
print(out["boxes"][keep], out["labels"][keep])
```

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- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: mAP, proposal recall, per-class AP, latency, IoU-threshold AP.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
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- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- medical detection
- defect localization
- document layout detection
- autonomous driving baselines
- research benchmarks

12. Common Mistakes

- wrong anchor labels
- not converting box coordinates
- ignoring proposal recall
- tiny dataset without augmentation
- unfair latency comparison with YOLO

13. Edge Cases / Limitations

- slower than YOLO
- memory-heavy proposals
- complex training
- small objects need feature pyramids

14. Variations

Item	Details
Fast R-CNN	external proposals
FPN Faster R-CNN	multi-scale features
Cascade R-CNN	staged high-IOU refinement
Mask R-CNN	adds masks

15. Related Topics

- Faster R-CNN vs YOLO: two-stage vs one-stage
- Faster R-CNN vs Mask R-CNN: boxes vs boxes plus masks
- RPN connects anchors and proposal learning

16. Interview Questions

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- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
- Resume value: Shows transfer learning and practical ML judgment.

19. Quick Revision

- Key idea: Faster R-CNN detects objects using a Region Proposal Network followed by a second-stage classifier and box regressor.
- Main formula: $t_x = (x - x_a) / w_a$, $t_y = (y - y_a) / h_a$.
- When to use: medical detection.
- Important metrics: mAP, proposal recall, per-class AP.
- Common traps: wrong anchor labels; not converting box coordinates.
- Interview one-liner: Faster R-CNN is a two-stage object detector used for medical detection and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	Faster R-CNN is a two-stage object detector.
Input/output	Image tensor plus ground-truth boxes/classes during training. -> Final boxes, class labels, and confidence scores.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	mAP, proposal recall, per-class AP, latency
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	medical detection, defect localization, document layout detection

Mask R-CNN

1. Overview

Mask R-CNN extends Faster R-CNN by adding a mask branch. It detects each object instance, classifies it, refines its box, and predicts the pixels belonging to that instance.

2. Intuition

Faster R-CNN draws rectangles. Mask R-CNN also colors the exact object pixels, even when two objects overlap.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes

- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Instance segmentation

- What it means: Separate object instances receive separate masks.
- Why it matters: It controls whether Mask R-CNN works correctly in training, evaluation, or deployment.
- Simple example: Two people are two masks, not one person blob.
- Common interview angle: How is it different from semantic segmentation?

Mask branch

- What it means: A small FCN predicts masks for each ROI.
- Why it matters: It controls whether Mask R-CNN works correctly in training, evaluation, or deployment.
- Simple example: A cat ROI gets a cat-shaped binary mask.
- Common interview angle: Why predict masks in parallel?

ROI Align

- What it means: Keeps spatial alignment for accurate masks.
- Why it matters: It controls whether Mask R-CNN works correctly in training, evaluation, or deployment.
- Simple example: Avoids shifted boundaries from quantized pooling.
- Common interview angle: Why did Mask R-CNN need ROI Align?

Multi-task loss

- What it means: Classification, box, and mask losses train together.
- Why it matters: It controls whether Mask R-CNN works correctly in training, evaluation, or deployment.
- Simple example: Shared features learn object identity and shape.
- Common interview angle: How do the losses combine?

5. Algorithm / Working Process

- Input: Image plus boxes, classes, and instance masks during training.
- Processing steps:
 1. Extract backbone features.
 2. Generate proposals using RPN.
 3. Crop aligned ROI features.
 4. Predict class and refined box.
 5. Predict mask for each ROI.
 6. Resize mask to the final image box.
- Output: Boxes, classes, scores, and one binary mask per detected instance.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $\text{total_loss} = L_{\text{cls}} + L_{\text{box}} + L_{\text{mask}}$
- L_{mask} = binary cross entropy over mask pixels
- $\text{Dice} = 2|P \cap G| / (|P| + |G|)$
- Mask AP averages mask IoU thresholds

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
import torch
from torchvision.models.detection import maskrcnn_resnet50_fpn

model = maskrcnn_resnet50_fpn(weights="DEFAULT").eval()
image = torch.rand(3, 512, 512)

with torch.no_grad():
    out = model([image])[0]

masks = out["masks"][:, 0] > 0.5
print(out["boxes"].shape, masks.shape)
```

8. Code Explanation

- The imports use the smallest common library for practice.
- The model block shows either a pretrained model or a tiny runnable version.
- The input block demonstrates the expected shape or file input.
- The output block prints the object you must understand in interviews: logits, boxes, masks, embeddings, generated image, or coordinates.
- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: mask AP, box AP, Dice, IoU, boundary quality.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.

- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- medical instance masks
- cell counting
- robot grasping
- product cutouts
- annotation pipelines

12. Common Mistakes

- using semantic masks as instance labels
- bad polygon conversion
- evaluating only boxes
- ignoring mask imbalance
- forgetting ROI Align

13. Edge Cases / Limitations

- slow
- requires instance masks
- coarse masks for thin objects
- crowded scenes are hard
- not promptable by default

14. Variations

Item	Details
Mask R-CNN + FPN	strong baseline
Cascade Mask R-CNN	better localization
YOLACT/SOLO	faster one-stage masks
SAM-style models	promptable masks

15. Related Topics

- Mask R-CNN vs U-Net: instance vs semantic segmentation
- Mask R-CNN vs Faster R-CNN: mask branch added
- Mask R-CNN vs SAM: class-aware training vs promptable foundation masks

16. Interview Questions

1. **What problem does it solve?** Answer: It solves the core task described in the overview and makes that task trainable, scalable, or deployable in real computer-vision systems.
2. **What is the input and output?** Answer: Input and output are listed in the working process; always answer with tensor/object shapes when possible.
3. **What is the main training signal?** Answer: The training signal is the relevant loss: cross-entropy, box loss, mask loss, contrastive loss, denoising loss, reconstruction loss, or language-modeling loss.
4. **Which metric should be used?** Answer: Choose the metric that matches the output, not generic accuracy.
5. **How do you debug bad results?** Answer: Visualize predictions, inspect labels, overfit a tiny batch, verify preprocessing, and check train/validation curves.
6. **What causes overfitting?** Answer: Small data, noisy labels, weak augmentation, too-large models, and validation leakage.
7. **What hyperparameters matter most?** Answer: Learning rate, batch size, resolution, model size, thresholds, augmentation, scheduler, and loss weights.
8. **What is the most common interview trap?** Answer: Only describing the architecture and forgetting loss, metrics, inference post-processing, and failure modes.
9. **How do you improve performance?** Answer: Clean data first, use pretrained weights, tune preprocessing and thresholds, then scale or modify architecture.
10. **How do you deploy it?** Answer: Export the model, reproduce preprocessing, benchmark latency/memory, monitor drift, and test on real edge cases.

17. Practice Tasks

- Small coding task: run the provided snippet and explain every tensor/output shape.
- Dataset-based project: train or evaluate on a tiny public/custom dataset.
- Experiment idea: compare pretrained vs randomly initialized behavior.
- Debugging task: break preprocessing or labels and identify the metric/visual symptom.
- Extension idea: add a visualization script that saves predictions for review.

18. Project Ideas

Mask R-CNN Mini Lab

- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
- Dataset suggestion: COCO subset, Oxford Pets, Cityscapes subset, medical masks, satellite images, or custom data
- Resume value: Shows implementation and evaluation ability.

Mask R-CNN Error Analysis Dashboard

- What it does: Save predictions, metrics, and failure examples.
- Tech stack: Python, Streamlit, pandas, OpenCV
- Dataset suggestion: A validation set with diverse hard examples
- Resume value: Strong resume signal because it shows engineering maturity.

Mask R-CNN Domain Adaptation Demo

- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
- Resume value: Shows transfer learning and practical ML judgment.

19. Quick Revision

- Key idea: Mask R-CNN extends Faster R-CNN by adding a mask branch.
- Main formula: $\text{total_loss} = L_{\text{cls}} + L_{\text{box}} + L_{\text{mask}}$.
- When to use: medical instance masks.
- Important metrics: mask AP, box AP, Dice.
- Common traps: using semantic masks as instance labels; bad polygon conversion.
- Interview one-liner: Mask R-CNN is a instance segmentation model used for medical instance masks and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	Mask R-CNN is a instance segmentation model.
Input/output	Image plus boxes, classes, and instance masks during training. -> Boxes, classes, scores, and one binary mask per detected instance.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	mask AP, box AP, Dice, IoU
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	medical instance masks, cell counting, robot grasping

U-Net

1. Overview

U-Net is an encoder-decoder CNN for pixel-wise segmentation. It is especially strong for biomedical imaging and small-data segmentation because skip connections preserve localization detail.

2. Intuition

The encoder understands context; the decoder rebuilds the mask. Skip connections copy fine details from early layers so boundaries are sharper.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Encoder

- What it means: Downsamples and learns semantic features.
- Why it matters: It controls whether U-Net works correctly in training, evaluation, or deployment.
- Simple example: Deep layers identify organs or roads.
- Common interview angle: Why downsample?

Decoder

- What it means: Upsamples features back to image resolution.
- Why it matters: It controls whether U-Net works correctly in training, evaluation, or deployment.
- Simple example: A 32x32 map becomes 256x256.
- Common interview angle: How is resolution restored?

Skip connections

- What it means: Concatenate encoder detail into decoder.
- Why it matters: It controls whether U-Net works correctly in training, evaluation, or deployment.
- Simple example: Edge details improve mask boundaries.
- Common interview angle: Why are skips essential?

Pixel loss

- What it means: Each pixel contributes to the objective.
- Why it matters: It controls whether U-Net works correctly in training, evaluation, or deployment.
- Simple example: Foreground/background BCE for binary masks.
- Common interview angle: Which metric handles imbalance?

5. Algorithm / Working Process

- Input: Image tensor.
- Processing steps:
 1. Normalize image and mask.
 2. Encode with conv and pooling blocks.
 3. Process bottleneck features.
 4. Upsample in decoder.
 5. Concatenate skip features.
 6. Output logits and train with pixel-level loss.
- Output: Pixel-wise class logits or binary mask.

- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $BCE = -[y \log(p) + (1-y)\log(1-p)]$
- $Dice = 2|P \text{ intersection } G|/(|P|+|G|)$
- $Dice \text{ loss} = 1 - Dice$
- $\text{output shape} = N \times C \times H \times W$

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
import torch
from torch import nn

class TinyUNet(nn.Module):
    def __init__(self, classes=1):
        super().__init__()
        self.enc1 = nn.Sequential(nn.Conv2d(3, 16, 3, padding=1), nn.ReLU(),
nn.Conv2d(16, 16, 3, padding=1), nn.ReLU())
        self.pool = nn.MaxPool2d(2)
        self.enc2 = nn.Sequential(nn.Conv2d(16, 32, 3, padding=1), nn.ReLU())
        self.up = nn.ConvTranspose2d(32, 16, 2, stride=2)
        self.dec = nn.Sequential(nn.Conv2d(32, 16, 3, padding=1), nn.ReLU(),
nn.Conv2d(16, classes, 1))

    def forward(self, x):
        skip = self.enc1(x)
        z = self.up(self.enc2(self.pool(skip)))
        return self.dec(torch.cat([z, skip], dim=1))

model = TinyUNet()
images = torch.randn(4, 3, 128, 128)
masks = torch.randint(0, 2, (4, 1, 128, 128)).float()
loss = nn.BCEWithLogitsLoss()(model(images), masks)
loss.backward()
```

8. Code Explanation

- The imports use the smallest common library for practice.
- The model block shows either a pretrained model or a tiny runnable version.
- The input block demonstrates the expected shape or file input.
- The output block prints the object you must understand in interviews: logits, boxes, masks, embeddings, generated image, or coordinates.

- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: Dice, IoU, pixel accuracy, foreground recall, boundary F1.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- tumor segmentation
- organ segmentation
- satellite road extraction
- industrial defect masks
- background removal

12. Common Mistakes

- using pixel accuracy on imbalanced masks
- misaligned augmentations
- wrong mask interpolation
- not checking boundaries
- too little augmentation

13. Edge Cases / Limitations

- limited global context
- needs pixel labels
- thin structures are hard
- can overfit small data

14. Variations

Item	Details
U-Net+ +	nested skips
Attention U-Net	gated skips
3D U-Net	volumetric data
TransUNet/Swin-UNet	Transformer hybrids

15. Related Topics

- U-Net vs Mask R-CNN: semantic vs instance masks
- U-Net vs FCN: stronger decoder and skips
- U-Net and Dice loss: common for imbalanced masks

16. Interview Questions

1. **What problem does it solve?** Answer: It solves the core task described in the overview and makes that task trainable, scalable, or deployable in real computer-vision systems.
2. **What is the input and output?** Answer: Input and output are listed in the working process; always answer with tensor/object shapes when possible.
3. **What is the main training signal?** Answer: The training signal is the relevant loss: cross-entropy, box loss, mask loss, contrastive loss, denoising loss, reconstruction loss, or language-modeling loss.
4. **Which metric should be used?** Answer: Choose the metric that matches the output, not generic accuracy.
5. **How do you debug bad results?** Answer: Visualize predictions, inspect labels, overfit a tiny batch, verify preprocessing, and check train/validation curves.
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10. **How do you deploy it?** Answer: Export the model, reproduce preprocessing, benchmark latency/memory, monitor drift, and test on real edge cases.

17. Practice Tasks

- Small coding task: run the provided snippet and explain every tensor/output shape.
- Dataset-based project: train or evaluate on a tiny public/custom dataset.
- Experiment idea: compare pretrained vs randomly initialized behavior.
- Debugging task: break preprocessing or labels and identify the metric/visual symptom.
- Extension idea: add a visualization script that saves predictions for review.

18. Project Ideas

U-Net Mini Lab

- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
- Dataset suggestion: COCO subset, Oxford Pets, Cityscapes subset, medical masks, satellite images, or custom data
- Resume value: Shows implementation and evaluation ability.

U-Net Error Analysis Dashboard

- What it does: Save predictions, metrics, and failure examples.
- Tech stack: Python, Streamlit, pandas, OpenCV
- Dataset suggestion: A validation set with diverse hard examples
- Resume value: Strong resume signal because it shows engineering maturity.

U-Net Domain Adaptation Demo

- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
- Resume value: Shows transfer learning and practical ML judgment.

19. Quick Revision

- Key idea: U-Net is an encoder-decoder CNN for pixel-wise segmentation.
- Main formula: $BCE = -[y \log(p) + (1-y)\log(1-p)]$.
- When to use: tumor segmentation.
- Important metrics: Dice, IoU, pixel accuracy.
- Common traps: using pixel accuracy on imbalanced masks; misaligned augmentations.
- Interview one-liner: U-Net is a semantic segmentation network used for tumor segmentation and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	U-Net is a semantic segmentation network.
Input/output	Image tensor. -> Pixel-wise class logits or binary mask.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	Dice, IoU, pixel accuracy, foreground recall
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	tumor segmentation, organ segmentation, satellite road extraction

Vision Transformer

1. Overview

A Vision Transformer, ViT, splits an image into patches and processes them as tokens with Transformer self-attention. It is used in classification, foundation backbones, detection, segmentation, and multimodal systems.

2. Intuition

A ViT reads an image like a sequence of patch tokens. Attention lets every patch directly compare itself with every other patch.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Patch embedding

- What it means: Flatten patches and project them to vectors.
- Why it matters: It controls whether Vision Transformer works correctly in training, evaluation, or deployment.
- Simple example: 224x224 with 16x16 patches gives 196 tokens.
- Common interview angle: Why patchify images?

Self-attention

- What it means: Tokens attend to other tokens.
- Why it matters: It controls whether Vision Transformer works correctly in training, evaluation, or deployment.
- Simple example: A wheel patch attends to the car body.
- Common interview angle: Why is attention expensive?

CLS token

- What it means: A learned summary token used for classification.
- Why it matters: It controls whether Vision Transformer works correctly in training, evaluation, or deployment.
- Simple example: CLS output feeds the classifier.
- Common interview angle: What does CLS represent?

Position embedding

- What it means: Adds spatial order.

- Why it matters: It controls whether Vision Transformer works correctly in training, evaluation, or deployment.
- Simple example: Patch sequence needs location information.
- Common interview angle: Why does ViT need positions?

5. Algorithm / Working Process

- Input: Image split into fixed-size patches.
- Processing steps:
 1. Resize image.
 2. Split into patches.
 3. Project patches to embeddings.
 4. Add position embeddings and optional CLS token.
 5. Run Transformer encoder blocks.
 6. Use CLS or pooled tokens for prediction.
- Output: Class logits, patch embeddings, or visual features.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$
- $N = H \cdot W / P^2$ patches
- attention cost = $O(N^2 d)$
- MLP block = Linear $\rightarrow$ GELU $\rightarrow$ Linear

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
import torch
from torchvision.models import vit_b_16, ViT_B_16_Weights

model = vit_b_16(weights=ViT_B_16_Weights.DEFAULT)
model.heads.head = torch.nn.Linear(model.heads.head.in_features, 5)

batch = torch.randn(2, 3, 224, 224)
print(model(batch).shape)  # [2, 5]
```

8. Code Explanation

- The imports use the smallest common library for practice.
- The model block shows either a pretrained model or a tiny runnable version.
- The input block demonstrates the expected shape or file input.

- The output block prints the object you must understand in interviews: logits, boxes, masks, embeddings, generated image, or coordinates.
- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: top-1 accuracy, top-5 accuracy, F1, throughput, calibration error.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- image classification
- foundation backbones
- medical classification
- remote sensing
- multimodal encoders

12. Common Mistakes

- training from scratch on tiny data
- wrong position embedding resize
- large patches miss small objects
- ignoring attention cost
- unfair CNN comparison

13. Edge Cases / Limitations

- data hungry
- quadratic attention
- weaker locality bias
- resolution changes need care

14. Variations

Item	Details
DeiT	data-efficient distillation
Swin Transformer	windowed hierarchical attention
MAE	masked image pretraining
Hybrid CNN-ViT	CNN stem plus Transformer

15. Related Topics

- CNN vs ViT: locality bias vs global attention
- ViT vs CLIP: architecture vs image-text pretraining
- ViT vs Swin: flat global tokens vs hierarchical windows

16. Interview Questions

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- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
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- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
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- Key idea: A Vision Transformer, ViT, splits an image into patches and processes them as tokens with Transformer self-attention.
- Main formula: $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$.
- When to use: image classification.
- Important metrics: top-1 accuracy, top-5 accuracy, F1.
- Common traps: training from scratch on tiny data; wrong position embedding resize.
- Interview one-liner: Vision Transformer is a Transformer-based vision backbone used for image classification and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	Vision Transformer is a Transformer-based vision backbone.
Input/output	Image split into fixed-size patches. -> Class logits, patch embeddings, or visual features.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	top-1 accuracy, top-5 accuracy, F1, throughput
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt

Item	Details
Best use cases	image classification, foundation backbones, medical classification

CLIP

1. Overview

CLIP learns aligned image and text embeddings from large image-caption pairs. It enables zero-shot classification, image search, text-image retrieval, dataset filtering, and multimodal model backbones.

2. Intuition

CLIP pulls matching image-text pairs together and pushes mismatched pairs apart. An image of a dog should be close to 'a photo of a dog'.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Dual encoder

- What it means: Separate encoders for image and text.
- Why it matters: It controls whether CLIP works correctly in training, evaluation, or deployment.
- Simple example: ViT encodes image; Transformer encodes prompt.
- Common interview angle: Why is retrieval efficient?

Contrastive loss

- What it means: Correct pairs are positives; other batch items are negatives.
- Why it matters: It controls whether CLIP works correctly in training, evaluation, or deployment.
- Simple example: Each image chooses its caption among the batch.
- Common interview angle: What are in-batch negatives?

Zero-shot prompts

- What it means: Class names become text prompts.
- Why it matters: It controls whether CLIP works correctly in training, evaluation, or deployment.
- Simple example: 'a photo of a cat' acts like a classifier weight.
- Common interview angle: Why does prompt wording matter?

Embedding space

- What it means: Cosine similarity compares modalities.
- Why it matters: It controls whether CLIP works correctly in training, evaluation, or deployment.
- Simple example: Text query retrieves matching images.
- Common interview angle: Why is CLIP useful without fine-tuning?

5. Algorithm / Working Process

- Input: Image and text prompt/caption.
- Processing steps:
 1. Collect image-text pairs.
 2. Encode images and texts.
 3. L2-normalize embeddings.
 4. Compute similarity matrix.
 5. Apply symmetric contrastive loss.
 6. At inference compare image embedding with prompt embeddings.
- Output: Comparable image and text embeddings plus similarity logits.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $s_{ij} = (I_i \cdot T_j) / \tau$
- $L_{\text{image}} = \text{CE}(s_i, \text{target}=i)$
- $L_{\text{text}} = \text{CE}(s_j, \text{target}=j)$
- $L = (L_{\text{image}} + L_{\text{text}}) / 2$

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
# pip install transformers pillow
from PIL import Image
from transformers import CLIPModel, CLIPProcessor

model = CLIPModel.from_pretrained("openai/clip-vit-base-patch32")
processor = CLIPProcessor.from_pretrained("openai/clip-vit-base-patch32")

image = Image.open("image.jpg").convert("RGB")
labels = ["a photo of a dog", "a photo of a car", "a photo of food"]
inputs = processor(text=labels, images=image, return_tensors="pt", padding=True)
probs = model(**inputs).logits_per_image.softmax(dim=1)
print(dict(zip(labels, probs[0].tolist())))
```

8. Code Explanation

- The imports use the smallest common library for practice.
- The model block shows either a pretrained model or a tiny runnable version.
- The input block demonstrates the expected shape or file input.
- The output block prints the object you must understand in interviews: logits, boxes, masks, embeddings, generated image, or coordinates.
- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: zero-shot accuracy, Recall@K, MRR, embedding quality, robustness.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- zero-shot classification
- semantic search
- dataset filtering
- text-guided generation
- retrieval

12. Common Mistakes

- poor prompts
- expecting exact counting
- ignoring web-data bias
- not normalizing embeddings
- expecting pixel masks

13. Edge Cases / Limitations

- weak spatial reasoning
- bias
- prompt sensitivity

- small details can be missed
- not generative alone

14. Variations

Item	Details
OpenCLIP	open-source CLIP training
SigLIP	sigmoid contrastive objective
ALIGN	large-scale dual encoder
CLIP adapters	domain adaptation

15. Related Topics

- CLIP vs ViT: CLIP may use ViT but adds text alignment
- CLIP vs captioning: retrieval vs generation
- CLIP and diffusion: text conditioning signal

16. Interview Questions

1. **What problem does it solve?** Answer: It solves the core task described in the overview and makes that task trainable, scalable, or deployable in real computer-vision systems.
2. **What is the input and output?** Answer: Input and output are listed in the working process; always answer with tensor/object shapes when possible.
3. **What is the main training signal?** Answer: The training signal is the relevant loss: cross-entropy, box loss, mask loss, contrastive loss, denoising loss, reconstruction loss, or language-modeling loss.
4. **Which metric should be used?** Answer: Choose the metric that matches the output, not generic accuracy.
5. **How do you debug bad results?** Answer: Visualize predictions, inspect labels, overfit a tiny batch, verify preprocessing, and check train/validation curves.
6. **What causes overfitting?** Answer: Small data, noisy labels, weak augmentation, too-large models, and validation leakage.
7. **What hyperparameters matter most?** Answer: Learning rate, batch size, resolution, model size, thresholds, augmentation, scheduler, and loss weights.
8. **What is the most common interview trap?** Answer: Only describing the architecture and forgetting loss, metrics, inference post-processing, and failure modes.
9. **How do you improve performance?** Answer: Clean data first, use pretrained weights, tune preprocessing and thresholds, then scale or modify architecture.
10. **How do you deploy it?** Answer: Export the model, reproduce preprocessing, benchmark latency/memory, monitor drift, and test on real edge cases.

17. Practice Tasks

- Small coding task: run the provided snippet and explain every tensor/output shape.
- Dataset-based project: train or evaluate on a tiny public/custom dataset.
- Experiment idea: compare pretrained vs randomly initialized behavior.
- Debugging task: break preprocessing or labels and identify the metric/visual symptom.

- Extension idea: add a visualization script that saves predictions for review.

18. Project Ideas

CLIP Mini Lab

- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
- Dataset suggestion: COCO subset, Oxford Pets, Cityscapes subset, medical masks, satellite images, or custom data
- Resume value: Shows implementation and evaluation ability.

CLIP Error Analysis Dashboard

- What it does: Save predictions, metrics, and failure examples.
- Tech stack: Python, Streamlit, pandas, OpenCV
- Dataset suggestion: A validation set with diverse hard examples
- Resume value: Strong resume signal because it shows engineering maturity.

CLIP Domain Adaptation Demo

- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
- Resume value: Shows transfer learning and practical ML judgment.

19. Quick Revision

- Key idea: CLIP learns aligned image and text embeddings from large image-caption pairs.
- Main formula: $s_{ij} = (\mathbf{I}_i \cdot \mathbf{T}_j) / \tau$.
- When to use: zero-shot classification.
- Important metrics: zero-shot accuracy, Recall@K, MRR.
- Common traps: poor prompts; expecting exact counting.
- Interview one-liner: CLIP is a contrastive image-text model used for zero-shot classification and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	CLIP is a contrastive image-text model.
Input/output	Image and text prompt/caption. -> Comparable image and text embeddings plus similarity logits.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	zero-shot accuracy, Recall@K, MRR, embedding quality

Item	Details
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	zero-shot classification, semantic search, dataset filtering

Diffusion for Vision

1. Overview

Diffusion models learn to reverse a gradual noising process. They power text-to-image generation, inpainting, image editing, super-resolution, denoising, and synthetic data generation.

2. Intuition

Training corrupts images with noise and teaches a model to remove that noise. Generation starts from random noise and repeatedly denoises it into an image.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Forward process

- What it means: A fixed schedule adds Gaussian noise.
- Why it matters: It controls whether Diffusion for Vision works correctly in training, evaluation, or deployment.
- Simple example: A clean cat image becomes pure noise.
- Common interview angle: Why is forward noising fixed?

Reverse process

- What it means: The model predicts how to denoise.
- Why it matters: It controls whether Diffusion for Vision works correctly in training, evaluation, or deployment.
- Simple example: Each step removes a little noise.
- Common interview angle: What does the network predict?

Conditioning

- What it means: Text or other signals guide denoising.
- Why it matters: It controls whether Diffusion for Vision works correctly in training, evaluation, or deployment.
- Simple example: A prompt steers image content.
- Common interview angle: How does classifier-free guidance work?

Latent diffusion

- What it means: Denoising happens in compressed VAE latent space.
- Why it matters: It controls whether Diffusion for Vision works correctly in training, evaluation, or deployment.
- Simple example: Stable Diffusion avoids pixel-space cost.
- Common interview angle: Why is it cheaper?

5. Algorithm / Working Process

- Input: Noise plus optional condition such as text, class, mask, edge map, or image.
- Processing steps:
 1. Sample image and timestep.
 2. Add scheduled noise.
 3. Predict noise/velocity/clean image.
 4. Optimize denoising loss.
 5. Start inference from random noise.
 6. Iteratively denoise and decode.
- Output: Generated, restored, edited, or enhanced image.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $q(x_t|x_0) = N(\sqrt{\alpha_t}x_0, (1-\alpha_t)\sigma_t^2)$
- $L = E[\epsilon - \epsilon_\theta(x_t, t, c)]^2$
- $\text{guidance} = \epsilon_{\text{uncond}} + w(\epsilon_{\text{cond}} - \epsilon_{\text{uncond}})$
- sampling repeats denoising over T or fewer steps

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
# pip install diffusers transformers accelerate torch
from diffusers import StableDiffusionPipeline
import torch

device = "cuda" if torch.cuda.is_available() else "cpu"
pipe = StableDiffusionPipeline.from_pretrained(
```

```
"runwayml/stable-diffusion-v1-5",
torch_dtype=torch.float16 if device == "cuda" else torch.float32,
).to(device)

image = pipe("a clean product photo of a blue running shoe",
num_inference_steps=25, guidance_scale=7.5).images[0]
image.save("generated_shoe.png")
```

8. Code Explanation

- The imports use the smallest common library for practice.
- The model block shows either a pretrained model or a tiny runnable version.
- The input block demonstrates the expected shape or file input.
- The output block prints the object you must understand in interviews: logits, boxes, masks, embeddings, generated image, or coordinates.
- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: FID, CLIP score, human preference, edit consistency, latency.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- text-to-image
- inpainting
- super-resolution
- synthetic data
- creative editing

12. Common Mistakes

- thinking generation is one step

- too high guidance
- ignoring licenses
- judging only cherry-picked samples
- confusing VAE latents and text embeddings

13. Edge Cases / Limitations

- slow sampling
- hallucination
- prompt sensitivity
- high training cost
- bias or memorization

14. Variations

Item	Details
DDPM	foundational diffusion
DDIM	faster sampling
Latent Diffusion	denoise compressed latents
ControlNet	condition on edges/pose/depth

15. Related Topics

- Diffusion vs GAN: iterative denoising vs adversarial training
- Diffusion and U-Net: U-Net denoiser is common
- Diffusion and CLIP: text embeddings guide generation

16. Interview Questions

1. **What problem does it solve?** Answer: It solves the core task described in the overview and makes that task trainable, scalable, or deployable in real computer-vision systems.
2. **What is the input and output?** Answer: Input and output are listed in the working process; always answer with tensor/object shapes when possible.
3. **What is the main training signal?** Answer: The training signal is the relevant loss: cross-entropy, box loss, mask loss, contrastive loss, denoising loss, reconstruction loss, or language-modeling loss.
4. **Which metric should be used?** Answer: Choose the metric that matches the output, not generic accuracy.
5. **How do you debug bad results?** Answer: Visualize predictions, inspect labels, overfit a tiny batch, verify preprocessing, and check train/validation curves.
6. **What causes overfitting?** Answer: Small data, noisy labels, weak augmentation, too-large models, and validation leakage.
7. **What hyperparameters matter most?** Answer: Learning rate, batch size, resolution, model size, thresholds, augmentation, scheduler, and loss weights.
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10. **How do you deploy it?** Answer: Export the model, reproduce preprocessing, benchmark latency/memory, monitor drift, and test on real edge cases.

17. Practice Tasks

- Small coding task: run the provided snippet and explain every tensor/output shape.
- Dataset-based project: train or evaluate on a tiny public/custom dataset.
- Experiment idea: compare pretrained vs randomly initialized behavior.
- Debugging task: break preprocessing or labels and identify the metric/visual symptom.
- Extension idea: add a visualization script that saves predictions for review.

18. Project Ideas

Diffusion for Vision Mini Lab

- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
- Dataset suggestion: COCO subset, Oxford Pets, Cityscapes subset, medical masks, satellite images, or custom data
- Resume value: Shows implementation and evaluation ability.

Diffusion for Vision Error Analysis Dashboard

- What it does: Save predictions, metrics, and failure examples.
- Tech stack: Python, Streamlit, pandas, OpenCV
- Dataset suggestion: A validation set with diverse hard examples
- Resume value: Strong resume signal because it shows engineering maturity.

Diffusion for Vision Domain Adaptation Demo

- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
- Resume value: Shows transfer learning and practical ML judgment.

19. Quick Revision

- Key idea: Diffusion models learn to reverse a gradual noising process.
- Main formula: $q(x_t|x_0)=N(\sqrt{\alpha_{\bar{t}}x_0,(1-\alpha_{\bar{t}})I)$.
- When to use: text-to-image.
- Important metrics: FID, CLIP score, human preference.
- Common traps: thinking generation is one step; too high guidance.
- Interview one-liner: Diffusion for Vision is a generative vision model used for text-to-image and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	Diffusion for Vision is a generative vision model.
Input/output	Noise plus optional condition such as text, class, mask, edge map, or image. -> Generated, restored, edited, or enhanced image.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	FID, CLIP score, human preference, edit consistency
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	text-to-image, inpainting, super-resolution

SAM-style Segmentation

1. Overview

SAM-style segmentation models predict masks from prompts such as points, boxes, or previous masks. They are used for interactive annotation, object cutouts, data labeling, and zero-shot segmentation workflows.

2. Intuition

Instead of training a new segmenter for every class, give the model a hint about the object and it returns likely masks.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Image encoder

- What it means: Computes dense visual embeddings.
- Why it matters: It controls whether SAM-style Segmentation works correctly in training, evaluation, or deployment.
- Simple example: Features can be cached for repeated prompts.
- Common interview angle: Why cache embeddings?

Prompt encoder

- What it means: Turns points/boxes/masks into prompt tokens.
- Why it matters: It controls whether SAM-style Segmentation works correctly in training, evaluation, or deployment.
- Simple example: A positive point marks foreground.
- Common interview angle: How does prompt type affect output?

Mask decoder

- What it means: Combines image and prompt embeddings to predict masks.
- Why it matters: It controls whether SAM-style Segmentation works correctly in training, evaluation, or deployment.
- Simple example: Ambiguous prompts can return multiple masks.
- Common interview angle: Why multiple outputs?

Quality score

- What it means: Predicts mask quality such as estimated IoU.
- Why it matters: It controls whether SAM-style Segmentation works correctly in training, evaluation, or deployment.
- Simple example: Choose the best mask automatically.
- Common interview angle: How do you rank masks?

5. Algorithm / Working Process

- Input: Image plus prompt: point, box, mask, or related cue.
- Processing steps:
 1. Encode image.
 2. Encode prompt.
 3. Decode candidate masks.
 4. Rank by predicted quality.
 5. Refine with more prompts if needed.
 6. Export selected mask.
- Output: One or more candidate masks and quality scores.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $\text{Dice} = 2 \frac{\text{sum}(pg)}{\text{sum } p + \text{sum } g}$
- $\text{IoU} = \frac{\text{intersection}}{\text{union}}$
- mask loss often uses BCE + Dice
- quality head estimates mask IoU

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
# pip install segment-anything opencv-python
import cv2
from segment_anything import sam_model_registry, SamPredictor

sam = sam_model_registry["vit_b"](checkpoint="sam_vit_b.pth")
predictor = SamPredictor(sam)

image = cv2.cvtColor(cv2.imread("image.jpg"), cv2.COLOR_BGR2RGB)
predictor.set_image(image)
masks, scores, _ = predictor.predict(
    point_coords=[[250, 180]],
    point_labels=[1],
)
print(masks.shape, scores)
```

8. Code Explanation

- The imports use the smallest common library for practice.
- The model block shows either a pretrained model or a tiny runnable version.
- The input block demonstrates the expected shape or file input.
- The output block prints the object you must understand in interviews: logits, boxes, masks, embeddings, generated image, or coordinates.
- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: mask IoU, Dice, boundary F1, prompt efficiency, latency.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- interactive annotation
- medical assistance
- background removal
- robotics segmentation
- dataset pre-labeling

12. Common Mistakes

- expecting class labels
- trusting ambiguous prompts
- ignoring domain shift
- not validating boundaries
- confusing promptable and semantic segmentation

13. Edge Cases / Limitations

- base model is not class-aware
- tiny/thin objects are hard
- prompt quality matters
- large encoders need memory
- specialized domains may need adaptation

14. Variations

Item	Details
SAM	original promptable model
MobileSAM/FastSAM	faster approximations
MedSAM	medical adaptation
SAM-style video segmentation	tracks masks through time

15. Related Topics

- SAM vs U-Net: promptable masks vs trained semantic masks
- SAM vs Mask R-CNN: prompt-driven vs class-aware instance masks
- SAM and active learning: faster labeling

16. Interview Questions

1. **What problem does it solve?** Answer: It solves the core task described in the overview and makes that task trainable, scalable, or deployable in real computer-vision systems.
2. **What is the input and output?** Answer: Input and output are listed in the working process; always answer with tensor/object shapes when possible.
3. **What is the main training signal?** Answer: The training signal is the relevant loss: cross-entropy, box loss, mask loss, contrastive loss, denoising loss, reconstruction loss, or language-modeling loss.
4. **Which metric should be used?** Answer: Choose the metric that matches the output, not generic accuracy.

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7. **What hyperparameters matter most?** Answer: Learning rate, batch size, resolution, model size, thresholds, augmentation, scheduler, and loss weights.
8. **What is the most common interview trap?** Answer: Only describing the architecture and forgetting loss, metrics, inference post-processing, and failure modes.
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- Small coding task: run the provided snippet and explain every tensor/output shape.
- Dataset-based project: train or evaluate on a tiny public/custom dataset.
- Experiment idea: compare pretrained vs randomly initialized behavior.
- Debugging task: break preprocessing or labels and identify the metric/visual symptom.
- Extension idea: add a visualization script that saves predictions for review.

18. Project Ideas

SAM-style Segmentation Mini Lab

- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
- Dataset suggestion: COCO subset, Oxford Pets, Cityscapes subset, medical masks, satellite images, or custom data
- Resume value: Shows implementation and evaluation ability.

SAM-style Segmentation Error Analysis Dashboard

- What it does: Save predictions, metrics, and failure examples.
- Tech stack: Python, Streamlit, pandas, OpenCV
- Dataset suggestion: A validation set with diverse hard examples
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SAM-style Segmentation Domain Adaptation Demo

- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
- Resume value: Shows transfer learning and practical ML judgment.

19. Quick Revision

- Key idea: SAM-style segmentation models predict masks from prompts such as points, boxes, or previous masks.

- Main formula: $\text{Dice} = 2 \cdot \text{sum}(pg) / (\text{sum } p + \text{sum } g)$.
- When to use: interactive annotation.
- Important metrics: mask IoU, Dice, boundary F1.
- Common traps: expecting class labels; trusting ambiguous prompts.
- Interview one-liner: SAM-style Segmentation is a promptable segmentation model used for interactive annotation and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	SAM-style Segmentation is a promptable segmentation model.
Input/output	Image plus prompt: point, box, mask, or related cue. -> One or more candidate masks and quality scores.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	mask IoU, Dice, boundary F1, prompt efficiency
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	interactive annotation, medical assistance, background removal

3D Vision

1. Overview

3D vision estimates depth, pose, shape, motion, and spatial structure from images, video, depth sensors, stereo cameras, or LiDAR. It is central to robotics, AR/VR, autonomous driving, mapping, and 3D reconstruction.

2. Intuition

2D vision says what appears in the image. 3D vision asks where it is in the world and how the scene is shaped.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Camera model

- What it means: Projects 3D points to 2D pixels.
- Why it matters: It controls whether 3D Vision works correctly in training, evaluation, or deployment.
- Simple example: World point becomes image coordinate using $K[R|t]$.
- Common interview angle: What are intrinsics and extrinsics?

Depth

- What it means: Distance from camera to scene point.
- Why it matters: It controls whether 3D Vision works correctly in training, evaluation, or deployment.
- Simple example: Every pixel can have a depth value.
- Common interview angle: Why is monocular depth ambiguous?

Point clouds

- What it means: Unordered 3D points from sensors or reconstruction.
- Why it matters: It controls whether 3D Vision works correctly in training, evaluation, or deployment.
- Simple example: LiDAR returns sparse car/road points.
- Common interview angle: Why are point clouds hard for CNNs?

3D detection

- What it means: Detects objects with 3D boxes.
- Why it matters: It controls whether 3D Vision works correctly in training, evaluation, or deployment.
- Simple example: Autonomous driving predicts car position and orientation.
- Common interview angle: How is 3D IoU different?

5. Algorithm / Working Process

- Input: Images, stereo pairs, RGB-D frames, LiDAR point clouds, or multi-view video.
- Processing steps:
 1. Calibrate sensors.
 2. Acquire images/depth/LiDAR.
 3. Transform data into common coordinates.
 4. Extract geometric representation.
 5. Run task model for depth/pose/detection/reconstruction.
 6. Evaluate with geometry-aware metrics.
- Output: Depth maps, camera poses, point clouds, meshes, 3D boxes, or reconstructed scenes.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $p \sim K[R|t]P$

- $Z = fB/d$ for stereo depth
- $X_c = R X_w + t$
- Chamfer distance compares point sets

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
import numpy as np

K = np.array([[800, 0, 320], [0, 800, 240], [0, 0, 1]], dtype=float)
R = np.eye(3)
t = np.array([0, 0, 2], dtype=float)
X_world = np.array([1.0, 0.5, 4.0])

X_cam = R @ X_world + t
x_h = K @ X_cam
pixel = (x_h[:2] / x_h[2]).ravel()
print(pixel)
```

8. Code Explanation

- The imports use the smallest common library for practice.
- The model block shows either a pretrained model or a tiny runnable version.
- The input block demonstrates the expected shape or file input.
- The output block prints the object you must understand in interviews: logits, boxes, masks, embeddings, generated image, or coordinates.
- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: AbsRel/RMSE depth, 3D IoU, Chamfer distance, pose error, trajectory error.
- Overfitting/underfitting: inspect both curves and qualitative predictions.
- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.

- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- autonomous driving
- robot navigation
- AR placement
- 3D scanning
- warehouse dimensioning

12. Common Mistakes

- mixing coordinate frames
- bad calibration
- using 2D metrics
- forgetting scale ambiguity
- not handling sparse data

13. Edge Cases / Limitations

- sensor noise
- occlusion
- monocular ambiguity
- voxel memory cost
- sensor domain shift

14. Variations

Item	Details
Stereo vision	two-camera depth
RGB-D vision	depth sensor plus RGB
PointNet	point-cloud network
NeRF/Gaussian splatting	neural scene representation

15. Related Topics

- 3D vs 2D vision: geometry vs image-plane labels
- Stereo vs monocular depth: calibrated geometry vs learned cues
- 3D vision and NeRF: scene representation

16. Interview Questions

1. **What problem does it solve?** Answer: It solves the core task described in the overview and makes that task trainable, scalable, or deployable in real computer-vision systems.
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17. Practice Tasks

- Small coding task: run the provided snippet and explain every tensor/output shape.
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- Experiment idea: compare pretrained vs randomly initialized behavior.
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- Extension idea: add a visualization script that saves predictions for review.

18. Project Ideas

3D Vision Mini Lab

- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
- Dataset suggestion: COCO subset, Oxford Pets, Cityscapes subset, medical masks, satellite images, or custom data
- Resume value: Shows implementation and evaluation ability.

3D Vision Error Analysis Dashboard

- What it does: Save predictions, metrics, and failure examples.
- Tech stack: Python, Streamlit, pandas, OpenCV
- Dataset suggestion: A validation set with diverse hard examples
- Resume value: Strong resume signal because it shows engineering maturity.

3D Vision Domain Adaptation Demo

- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
- Resume value: Shows transfer learning and practical ML judgment.

19. Quick Revision

- Key idea: 3D vision estimates depth, pose, shape, motion, and spatial structure from images, video, depth sensors, stereo cameras, or LiDAR.
- Main formula: $p \sim K[R|t]P$.
- When to use: autonomous driving.
- Important metrics: AbsRel/RMSE depth, 3D IoU, Chamfer distance.
- Common traps: mixing coordinate frames; bad calibration.
- Interview one-liner: 3D Vision is a geometry-aware computer vision used for autonomous driving and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	3D Vision is a geometry-aware computer vision.
Input/output	Images, stereo pairs, RGB-D frames, LiDAR point clouds, or multi-view video. -> Depth maps, camera poses, point clouds, meshes, 3D boxes, or reconstructed scenes.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	AbsRel/RMSE depth, 3D IoU, Chamfer distance, pose error
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	autonomous driving, robot navigation, AR placement

NeRF

1. Overview

NeRF, Neural Radiance Fields, represents a 3D scene as a neural function mapping 3D position and viewing direction to density and color. It synthesizes novel views from posed images.

2. Intuition

Instead of storing a mesh, NeRF learns a continuous field. A camera ray samples many points, queries color/density, and volume-renders them into a pixel.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes

- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Radiance field

- What it means: Function returns color and density.
- Why it matters: It controls whether NeRF works correctly in training, evaluation, or deployment.
- Simple example: A chair surface has high density and chair color.
- Common interview angle: What is density?

Volume rendering

- What it means: Accumulates color along camera rays.
- Why it matters: It controls whether NeRF works correctly in training, evaluation, or deployment.
- Simple example: Empty space contributes little opacity.
- Common interview angle: How is a pixel rendered?

Positional encoding

- What it means: Adds high-frequency coordinate features.
- Why it matters: It controls whether NeRF works correctly in training, evaluation, or deployment.
- Simple example: Helps model texture and sharp edges.
- Common interview angle: Why does plain MLP blur?

Novel view synthesis

- What it means: Renders unseen camera positions.
- Why it matters: It controls whether NeRF works correctly in training, evaluation, or deployment.
- Simple example: Move around a captured object virtually.
- Common interview angle: What data is required?

5. Algorithm / Working Process

- Input: Multiple images with known or estimated camera poses.
- Processing steps:
 1. Collect posed images.
 2. Cast rays through training pixels.
 3. Sample points along each ray.
 4. Predict color and density with MLP.
 5. Volume-render predicted pixel color.
 6. Optimize photometric reconstruction loss.
- Output: Rendered novel views and an implicit 3D scene field.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- $F_{\theta}(x, d) \rightarrow (\sigma, c)$
- $C(r) = \sum_i T_i (1 - \exp(-\sigma_i \delta_i)) c_i$
- $T_i = \exp(-\sum_{j < i} \sigma_j \delta_j)$
- $\text{loss} = \sum \|C_{\hat{r}}(r) - C(r)\|^2$

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
import torch
from torch import nn

class TinyRadianceField(nn.Module):
    def __init__(self):
        super().__init__()
        self.net = nn.Sequential(nn.Linear(6, 128), nn.ReLU(), nn.Linear(128,
128), nn.ReLU())
        self.sigma = nn.Linear(128, 1)
        self.rgb = nn.Sequential(nn.Linear(128, 3), nn.Sigmoid())

    def forward(self, xyz, direction):
        h = self.net(torch.cat([xyz, direction], dim=-1))
        return torch.relu(self.sigma(h)), self.rgb(h)

field = TinyRadianceField()
sigma, rgb = field(torch.randn(1024, 3), torch.randn(1024, 3))
print(sigma.shape, rgb.shape)
```

8. Code Explanation

- The imports use the smallest common library for practice.
- The model block shows either a pretrained model or a tiny runnable version.
- The input block demonstrates the expected shape or file input.
- The output block prints the object you must understand in interviews: logits, boxes, masks, embeddings, generated image, or coordinates.
- For projects, wrap this into a dataset class, training loop, validation loop, and visualization script.

9. Training / Evaluation

- Dataset preparation: inspect labels visually, normalize consistently, and avoid leakage across similar scenes, videos, patients, or identities.
- Train/validation/test split: validation tunes choices; test is used once for final reporting.
- Metrics: PSNR, SSIM, LPIPS, render time, memory.
- Overfitting/underfitting: inspect both curves and qualitative predictions.

- Hyperparameters: learning rate, batch size, resolution, augmentation, optimizer, scheduler, thresholds, and loss weights.
- Improve performance by cleaning labels, using pretrained weights, adding targeted augmentation, balancing classes, and doing error analysis.

10. Complexity and Cost

- Training cost grows with image resolution, model size, dataset size, and dense outputs.
- Inference cost depends on backbone size, number of tokens/proposals/pixels, and post-processing.
- Memory is dominated by activations during training and by feature maps/tokens at inference.
- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- novel view synthesis
- 3D asset capture
- AR/VR scenes
- robot simulation
- heritage digitization

12. Common Mistakes

- bad camera poses
- too few views
- expecting vanilla NeRF to be fast
- changing lighting
- confusing NeRF with mesh reconstruction

13. Edge Cases / Limitations

- slow vanilla training
- needs posed views
- dynamic scenes are harder
- poor extrapolation
- geometry is implicit

14. Variations

Item	Details
Instant-NGP	hash-grid acceleration
Mip-NeRF	anti-aliasing
Dynamic NeRF	time-varying scenes
3D Gaussian Splatting	fast explicit primitives

15. Related Topics

- NeRF vs mesh: implicit field vs explicit surface
- NeRF vs Gaussian splatting: neural volume vs fast splats
- NeRF and SLAM: camera poses matter

16. Interview Questions

1. **What problem does it solve?** Answer: It solves the core task described in the overview and makes that task trainable, scalable, or deployable in real computer-vision systems.
2. **What is the input and output?** Answer: Input and output are listed in the working process; always answer with tensor/object shapes when possible.
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- What it does: Build a small reproducible notebook around the topic.
- Tech stack: PyTorch/torchvision/OpenCV or the shown library
- Dataset suggestion: COCO subset, Oxford Pets, Cityscapes subset, medical masks, satellite images, or custom data
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- Tech stack: Python, Streamlit, pandas, OpenCV
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NeRF Domain Adaptation Demo

- What it does: Apply a pretrained model to a new domain and improve it.
- Tech stack: PyTorch, augmentation, experiment tracking
- Dataset suggestion: Traffic, retail, medical, document, or drone images
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- Key idea: NeRF, Neural Radiance Fields, represents a 3D scene as a neural function mapping 3D position and viewing direction to density and color.
- Main formula: $F_{\theta}(x,d) \rightarrow (\sigma, c)$.
- When to use: novel view synthesis.
- Important metrics: PSNR, SSIM, LPIPS.
- Common traps: bad camera poses; too few views.
- Interview one-liner: NeRF is a neural scene representation used for novel view synthesis and related vision systems.

20. Final Cheat Sheet

Item	Details
Definition	NeRF is a neural scene representation.
Input/output	Multiple images with known or estimated camera poses. -> Rendered novel views and an implicit 3D scene field.
Main steps	preprocess -> model forward -> decode/post-process -> evaluate
Key hyperparameters	learning rate, batch size, resolution, model size, thresholds/loss weights
Metrics	PSNR, SSIM, LPIPS, render time
Pros	practical, interview-relevant, supported by pretrained models or clear implementations
Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt
Best use cases	novel view synthesis, 3D asset capture, AR/VR scenes

Multimodal Vision-Language Models

1. Overview

Multimodal vision-language models combine image understanding with language generation or reasoning. They support captioning, VQA, document AI, chart understanding, grounding, OCR reasoning, and image chat.

2. Intuition

The model looks at an image, converts visual content into tokens or embeddings, and lets a language model answer using those visual signals.

3. Prerequisites

- linear algebra/probability basics
- PyTorch tensor shapes
- CNN/Transformer fundamentals
- train/validation/test discipline
- task-specific metrics

4. Core Concepts

Vision encoder

- What it means: Converts image to visual embeddings.
- Why it matters: It controls whether Multimodal Vision-Language Models works correctly in training, evaluation, or deployment.
- Simple example: ViT patches become tokens.
- Common interview angle: Why not feed raw pixels into an LLM?

Connector

- What it means: Projects visual features to language-model space.
- Why it matters: It controls whether Multimodal Vision-Language Models works correctly in training, evaluation, or deployment.
- Simple example: An MLP maps ViT dimension to LLM embedding dimension.
- Common interview angle: What does the projection layer do?

Cross-modal attention

- What it means: Text interacts with visual tokens.
- Why it matters: It controls whether Multimodal Vision-Language Models works correctly in training, evaluation, or deployment.
- Simple example: Question tokens attend to image regions.
- Common interview angle: How does VQA use attention?

Instruction tuning

- What it means: Trains the model to follow image-text instructions.
- Why it matters: It controls whether Multimodal Vision-Language Models works correctly in training, evaluation, or deployment.
- Simple example: Question-answer pairs teach useful responses.
- Common interview angle: Why is pretraining not enough?

5. Algorithm / Working Process

- Input: Image plus text prompt/question.
- Processing steps:
 1. Preprocess image.
 2. Encode image features.
 3. Project visual embeddings.
 4. Combine with text prompt.
 5. Decode response autoregressively.
 6. Optionally fine-tune on instruction data.
- Output: Answer, caption, reasoning text, grounded boxes/masks, or retrieval result.
- Training process: optimize the listed loss on a clean split, usually starting from pretrained weights when available.
- Inference process: run preprocessing, forward pass, decoding/post-processing, thresholding if relevant, and metric/visual checks.

6. Mathematical Foundation

- autoregressive loss = $-\sum_t \log p(y_t | y_{<t}, \text{image}, \text{prompt})$
- attention = $\text{softmax}(QK^T / \sqrt{d})V$
- contrastive alignment may use CLIP loss
- grounding adds box/mask losses when available

Loss functions should be explained as: what is predicted, what is ground truth, what error is penalized, and how it affects gradients.

7. Practical Implementation

```
# pip install transformers pillow
from PIL import Image
from transformers import BlipProcessor, BlipForQuestionAnswering

processor = BlipProcessor.from_pretrained("Salesforce/blip-vqa-base")
model = BlipForQuestionAnswering.from_pretrained("Salesforce/blip-vqa-base")

image = Image.open("image.jpg").convert("RGB")
inputs = processor(image, "What object is on the table?", return_tensors="pt")
tokens = model.generate(**inputs, max_new_tokens=20)
print(processor.decode(tokens[0], skip_special_tokens=True))
```

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- GPU is usually required for training; CPU can handle only small demos or offline inference.
- Always benchmark on target hardware before claiming real-time performance.

11. Common Use Cases

- visual question answering
- captioning
- document AI
- medical assistants
- robot instruction following

12. Common Mistakes

- trusting hallucinated explanations
- ignoring OCR resolution
- using only text metrics
- poor instruction data
- not testing ambiguous images

13. Edge Cases / Limitations

- hallucination
- weak counting
- high inference cost
- resolution sensitivity
- subjective evaluation

14. Variations

Item	Details
CLIP-style dual encoders	retrieval
BLIP/BLIP-2	vision-language pretraining
LLaVA-style	visual chat
Grounded VLMs	boxes/masks/citations

15. Related Topics

- VLM vs CLIP: generative chat vs embedding alignment
- VLM vs OCR pipeline: end-to-end reasoning vs specialized extraction
- VLM and SAM: language can select, SAM can segment

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- Important metrics: VQA accuracy, CIDEr/BLEU, grounding IoU.
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Cons	data quality, compute cost, domain shift, and evaluation mismatch can hurt

Item	Details
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